

A Statistical Toolkit for Linguistic Typology: Semantic maps

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Plan (tentative)

- Session 1: testing cross-linguistic generalizations
- **Session 2: creating semantic maps**
- Session 3:
 - comparing constraints
 - discussing your own ideas

Course materials

See the slides, data and code at
<https://github.com/levshina/Frejus2026>

Session 2

Outline

- Connectivity maps
 - Main principles
 - A hands-on case study with R: colexifications of FOOT
- Proximity maps
 - Main principles
 - A hands-on case study with R: European causatives

Overviews

- Georgakopoulos, T., & Polis, S. (2018). The semantic map model: State of the art and future avenues for linguistic research. *Language and Linguistics Compass* 12: e12270. <https://doi.org/10.1111/lnc3.12270>
- Haspelmath, M. (2003). The geometry of grammatical meaning: Semantic maps and cross-linguistic comparison. In M. Tomasello (Ed.), *The new psychology of language*, vol. 2 (pp. 211-242). Lawrence Erlbaum.
- Levshina, N. 2017. SÉMANTICKÁ MAPA (Semantic maps). In P. Karlík, M. Nekula, J. Pleskalová (eds.) *Nový encyklopedický slovník češtiny* (New Encyclopedic Dictionary of Czech). Nakladatelství Lidové noviny. English version: https://levshina.github.io/docs/Levshina2017_SemanticMaps.pdf
- van der Auwera, J. (2013). Semantic maps, for synchronic and diachronic typology. In A. Giacalone Ramat, C. Mauri & P. Molinelli (Eds.), *Synchrony and diachrony: a dynamic interface* (pp. 153-176). Benjamins.

Semantics and space

- Semantic **fields**
- **Near**-synonymy
- semantic **contiguity**
- Vector **space** models
- **cosine** measures of similarity
- **broadening** and **narrowing**
- ...

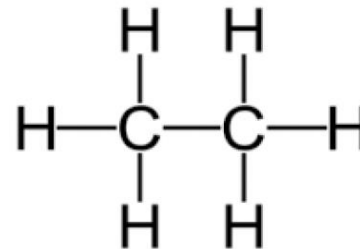
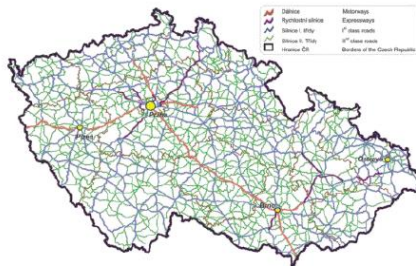
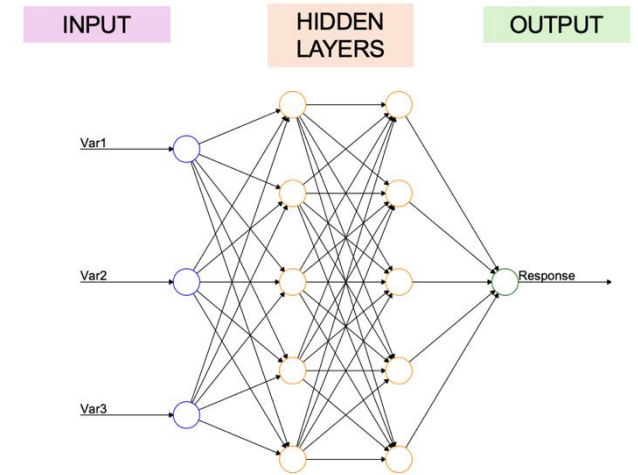
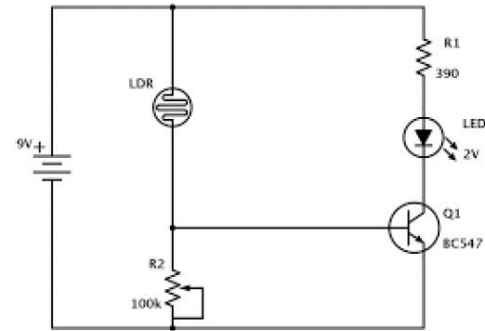
Semantic maps

- Represent a convenient tool for comparison of semantic and pragmatic functions across languages
- Based on different kinds of data:
 - grammars and typological databases,
 - dictionaries
 - parallel corpora
 - experimental data

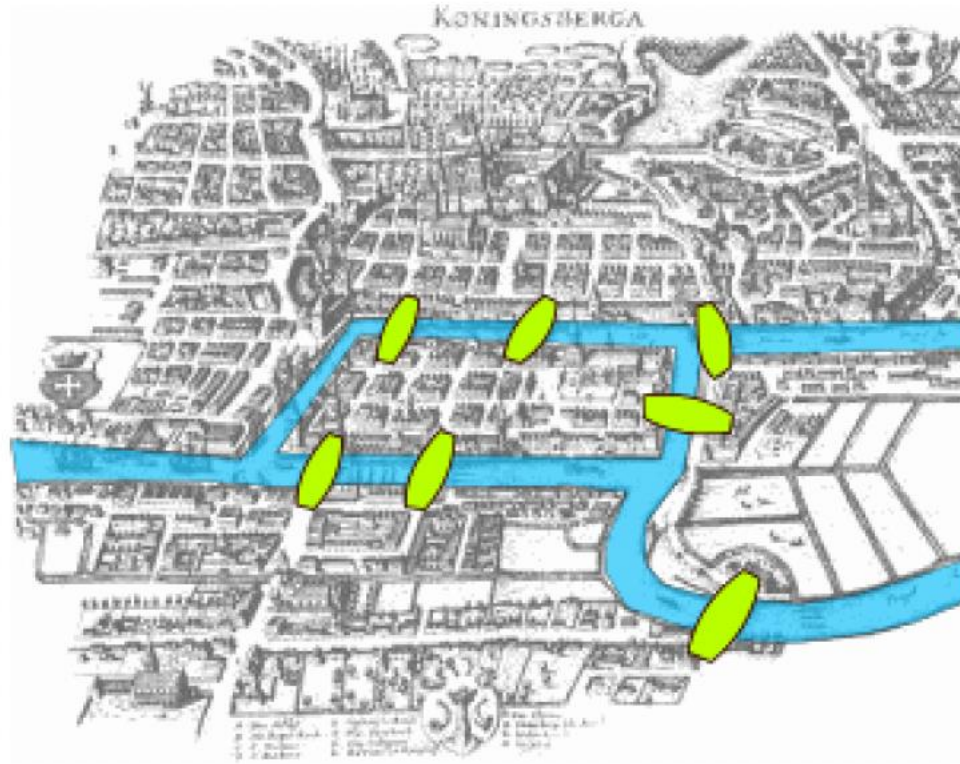
Main types of semantic maps

- **Connectivity maps** are graphs, where meanings are nodes (vertices) connected by links (edges).
 - Undirected vs. directed (e.g. diachrony)
 - Weighted vs. unweighted
- **Probabilistic maps**, with distances between objects (e.g. examples from a parallel corpus or stimuli in an experiment)

Graphs are everywhere!



Seven bridges of Königsberg

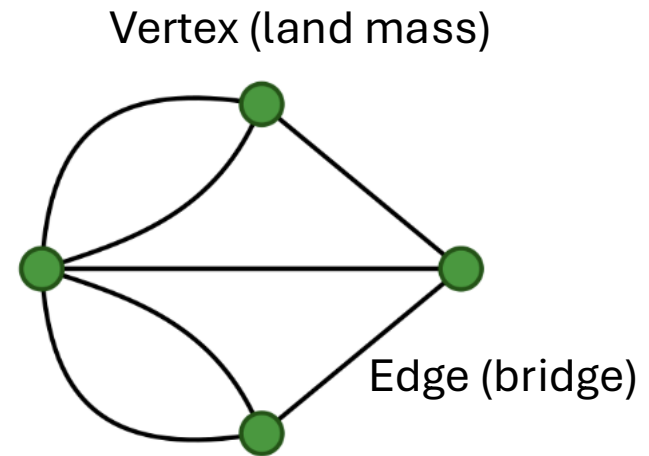


Devise a walk through the city that would cross each of the bridges once, and only once.

Graph theory



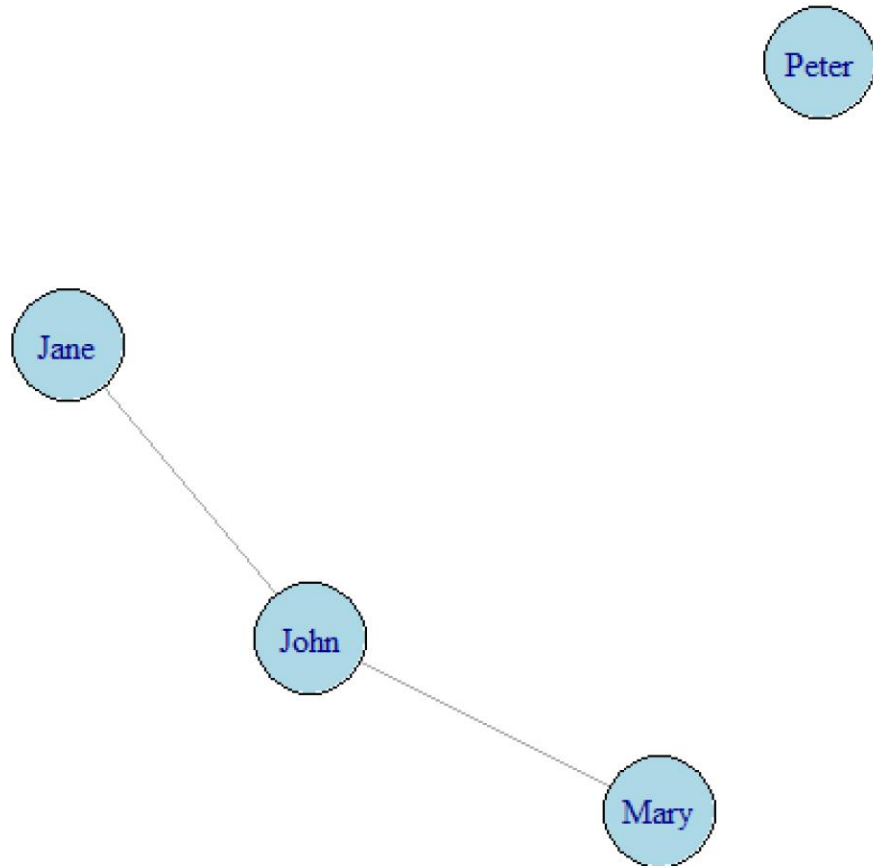
Leonhard Euler (1707 -1783)



Main types of graphs

- Undirected
 - Directed
-
- Unweighted
 - Weighted

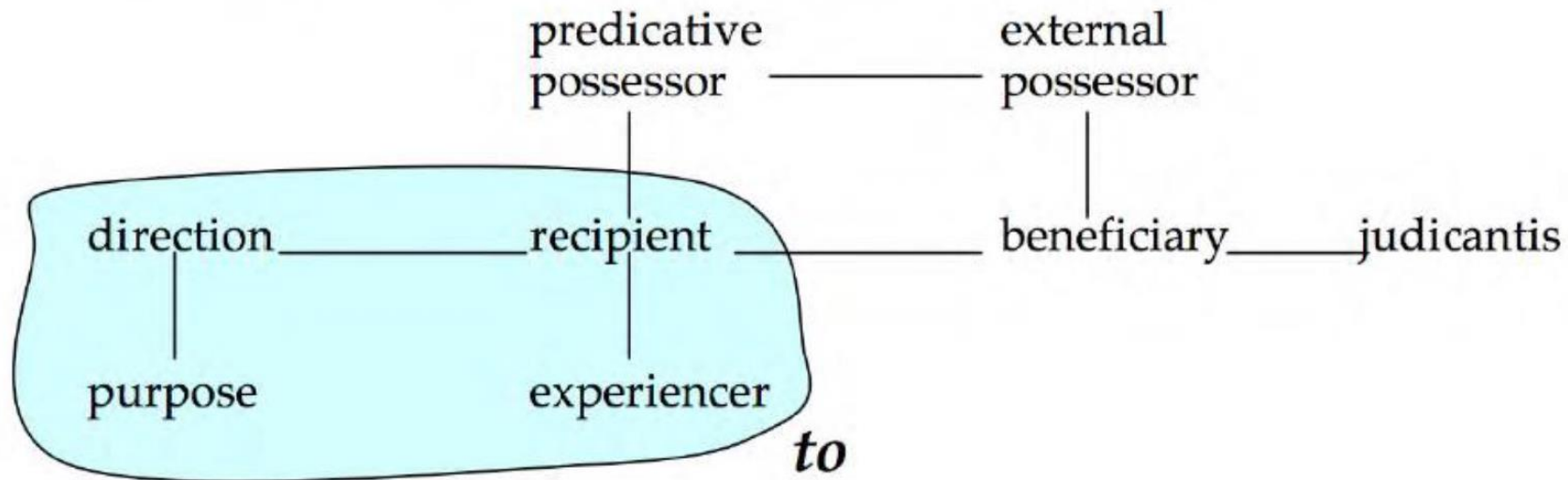
Undirected graphs



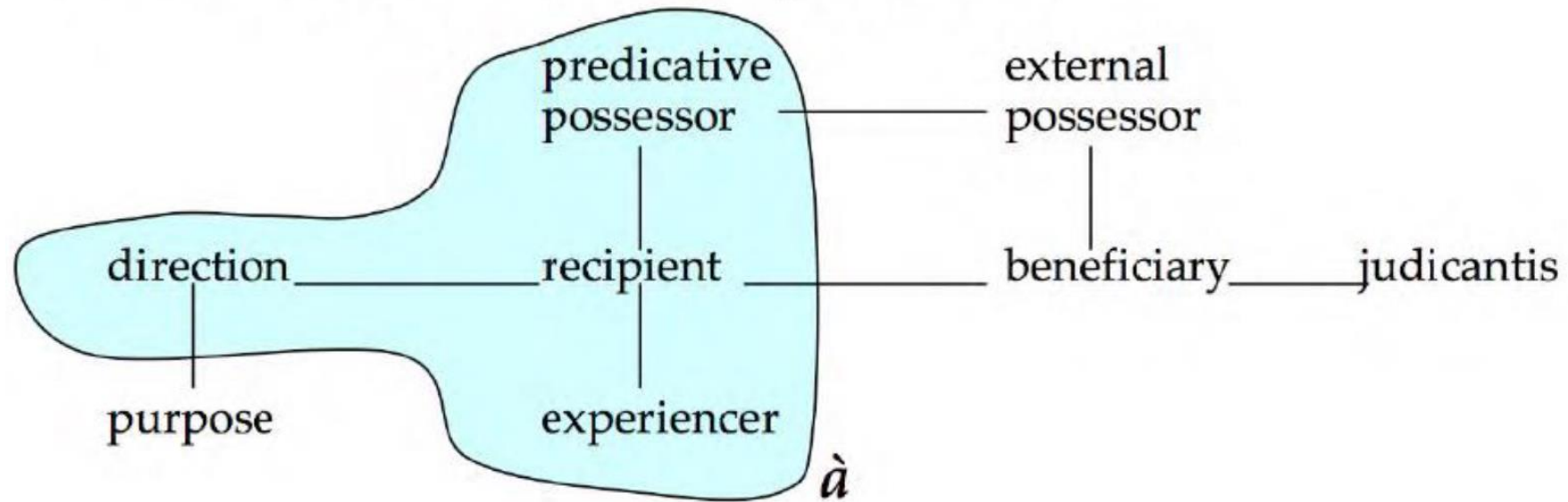
- Peter is single
- John and Mary are married.
- Jane is John's ex-wife.

Haspelmath (2003): datives

the English **Dative** preposition *to*



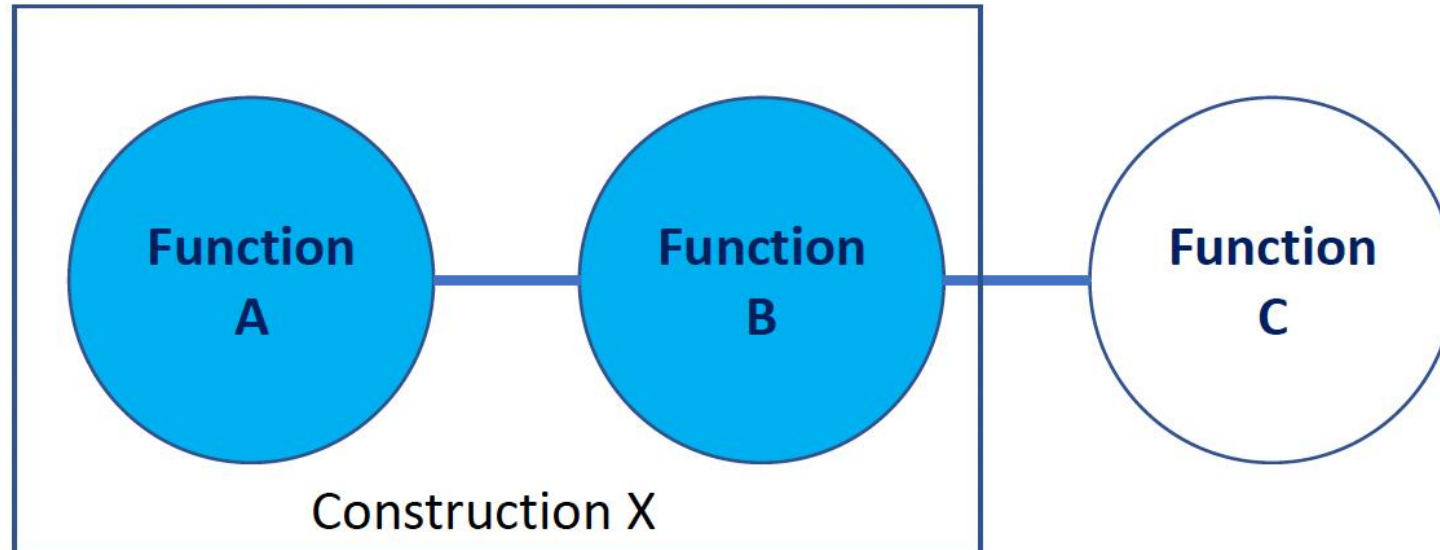
the French **Dative** preposition *à*

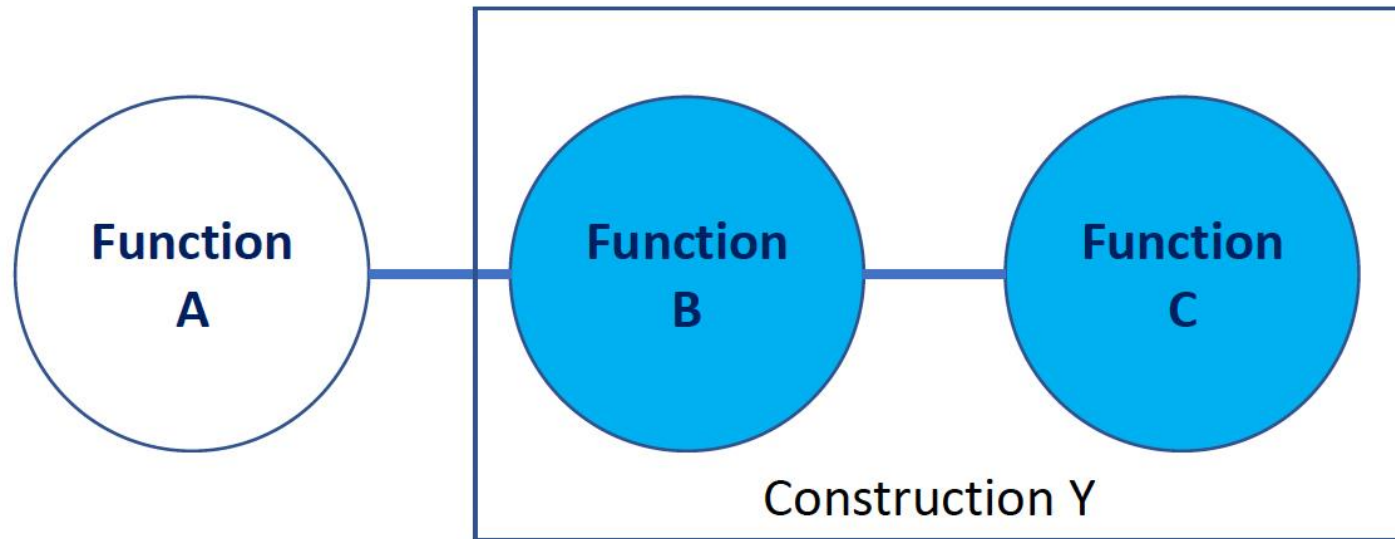


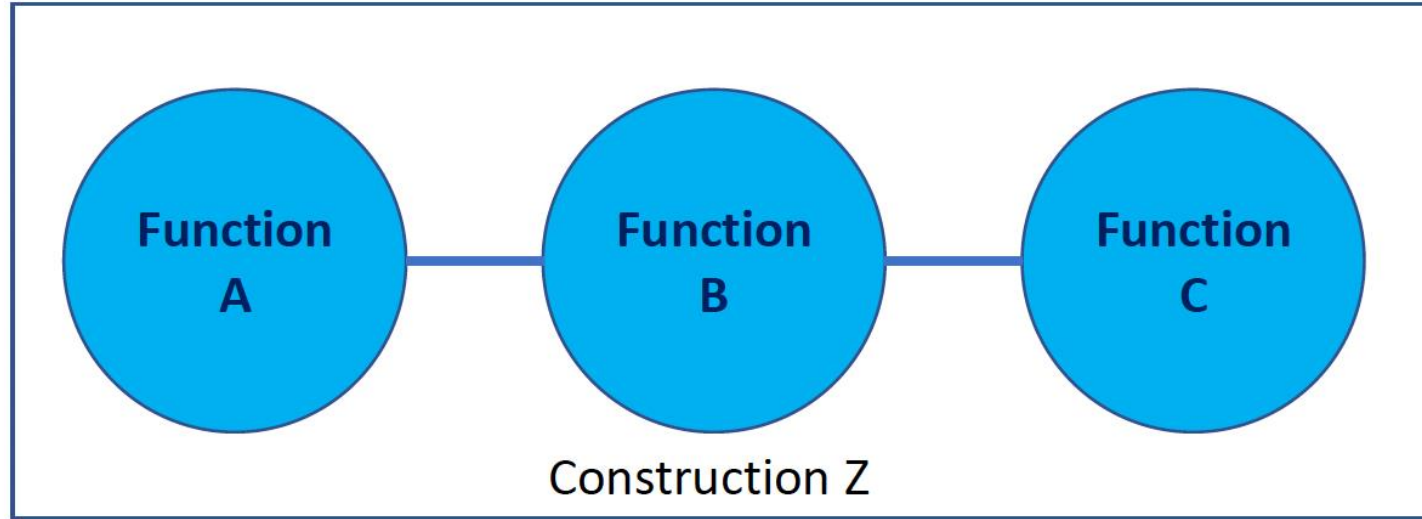
Main principles of connectivity maps

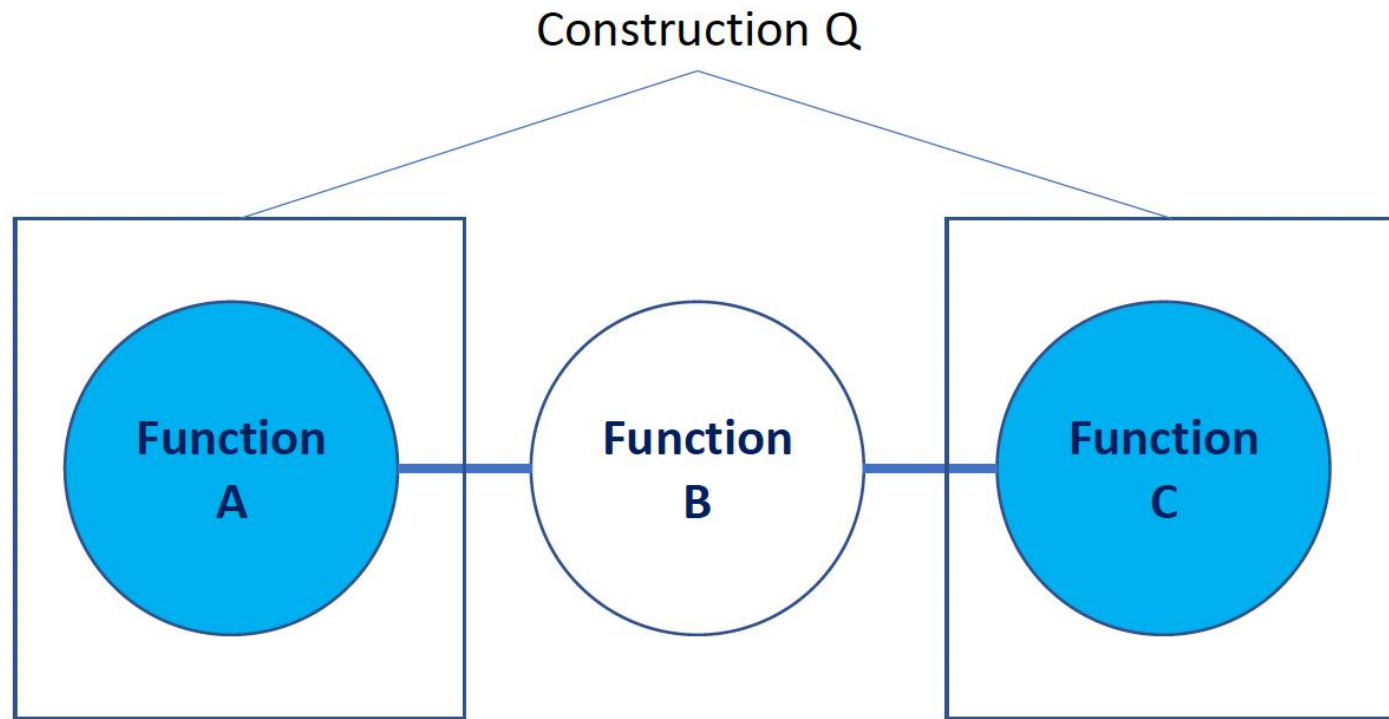
- Nodes (vertices): A function is put on a map when there's at least one pair of languages which differ wrt. this function (Haspelmath 2003)
- Links (edges): the principle of connectivity (adjacency/contiguity): if a construction has more than one function, they should be connected (see van der Auwera 2013).

Imaginary semantic map



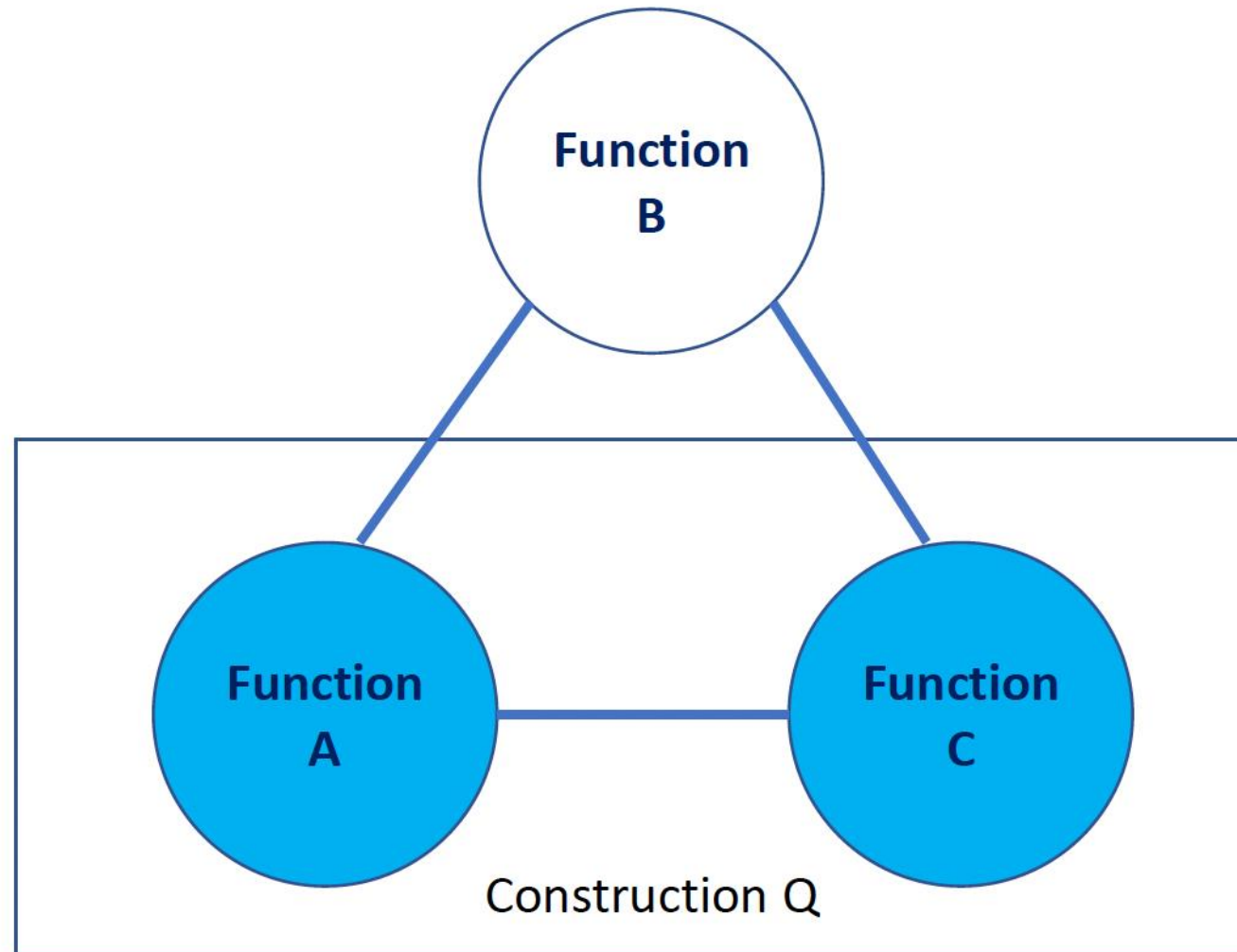






Wrong:
the connectivity principle is not observed!

A fix



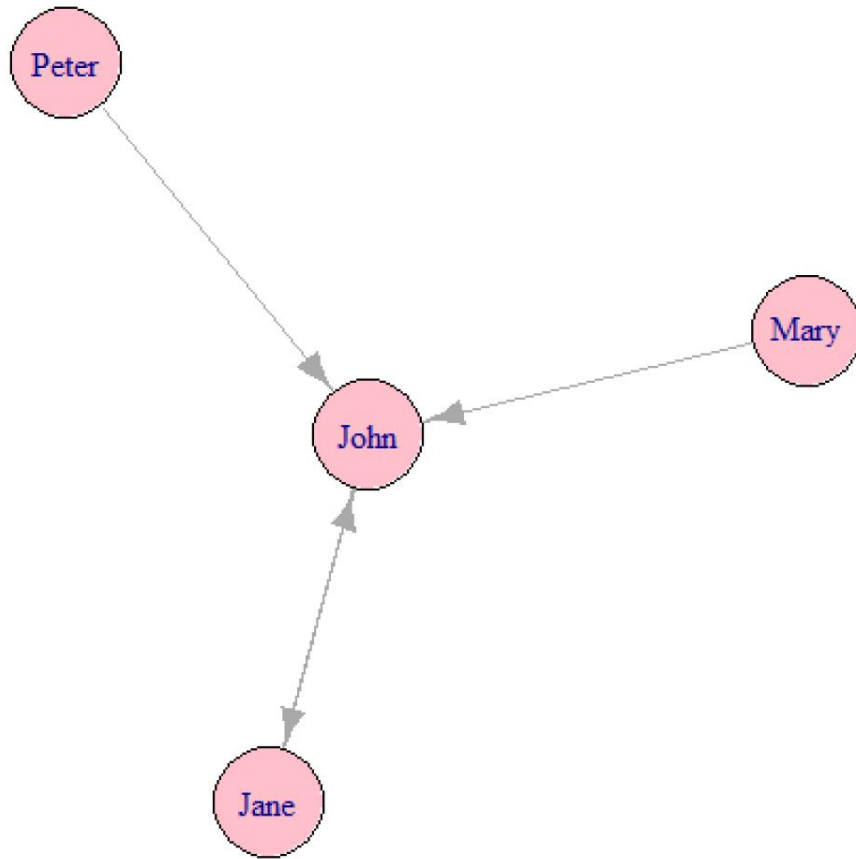
Parsimony

- "We will consider a map to be high-quality if it is relatively parsimonious – i.e., if it accommodates the data using few edges..." (Regier et al. 2013)
- A theoretical model: hypotheses about paths of semantic change

Controversy: cognitive status

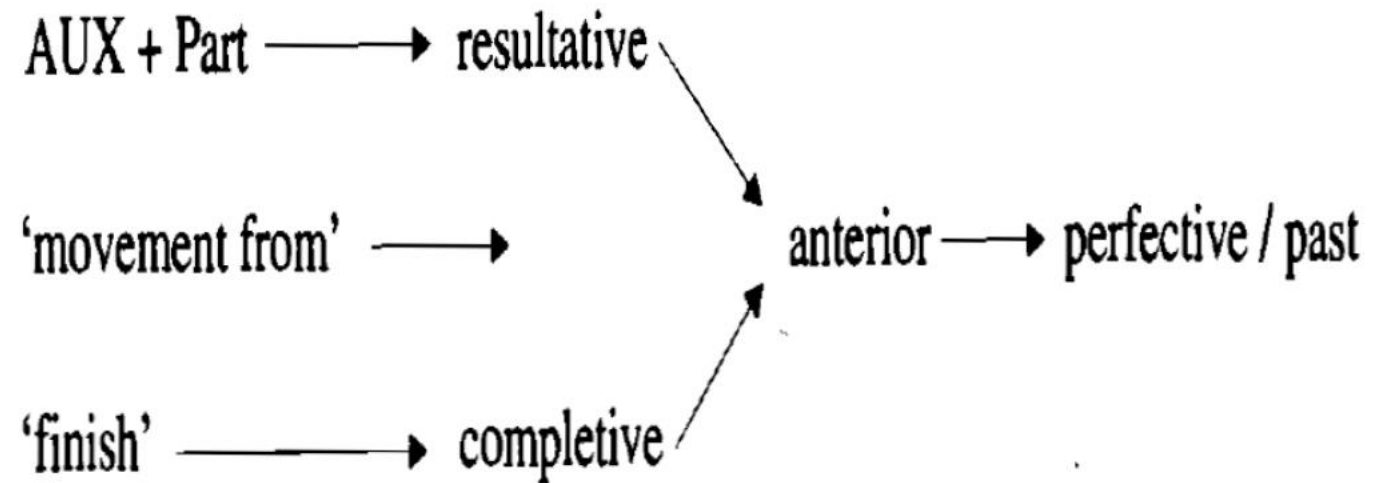
- A semantic map reflects universal organization of conceptual knowledge across languages, a 'geography of the human mind' (Croft 2001)
- Scepticism (Cristofaro 2010)
- How to test?

Directed graphs



- Mary loves John.
- John loves Jane.
- Jane loves John.
- And Peter fancies John, too.

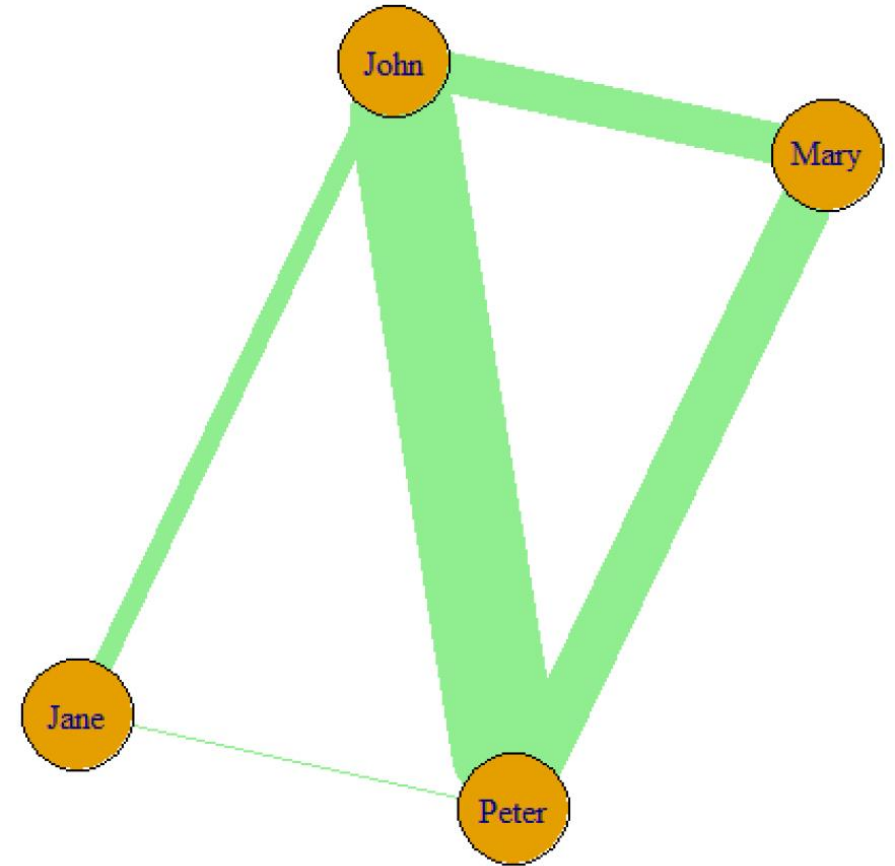
Tense and Aspect Grams



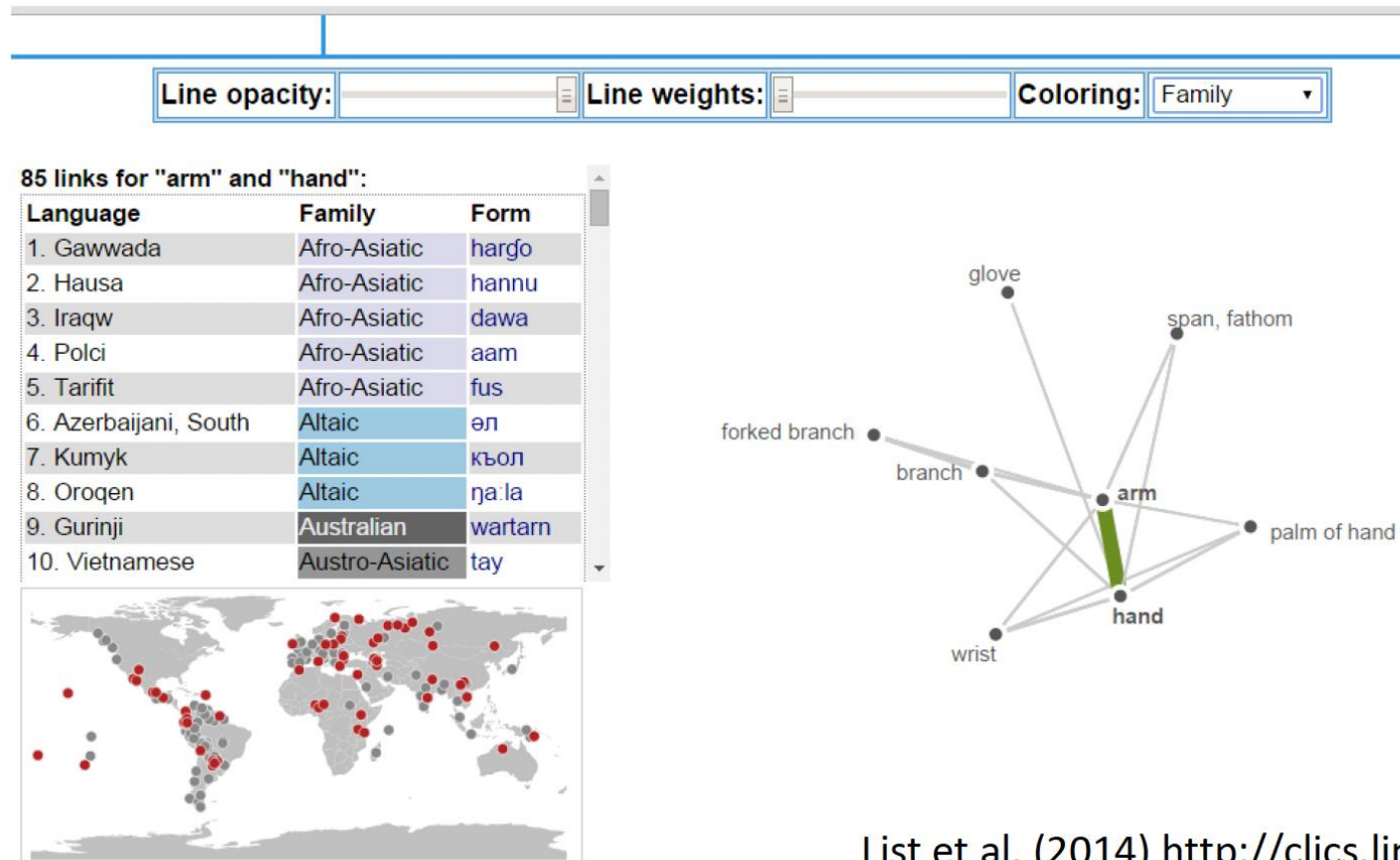
Bybee et al. 1994

Weighted graphs

- Mary and John have eaten 20 pizzas.
- John and Jane have eaten 10 pizzas.
- Peter and John have eaten 50 pizzas.
- Mary and Peter have eaten 25 pizzas.
- Jane and Mary have eaten 0 pizzas.
- Jane and Peter have eaten 1 pizza.



Colexification patterns of HAND



List et al. (2014) <http://clics.lingpy.org/>

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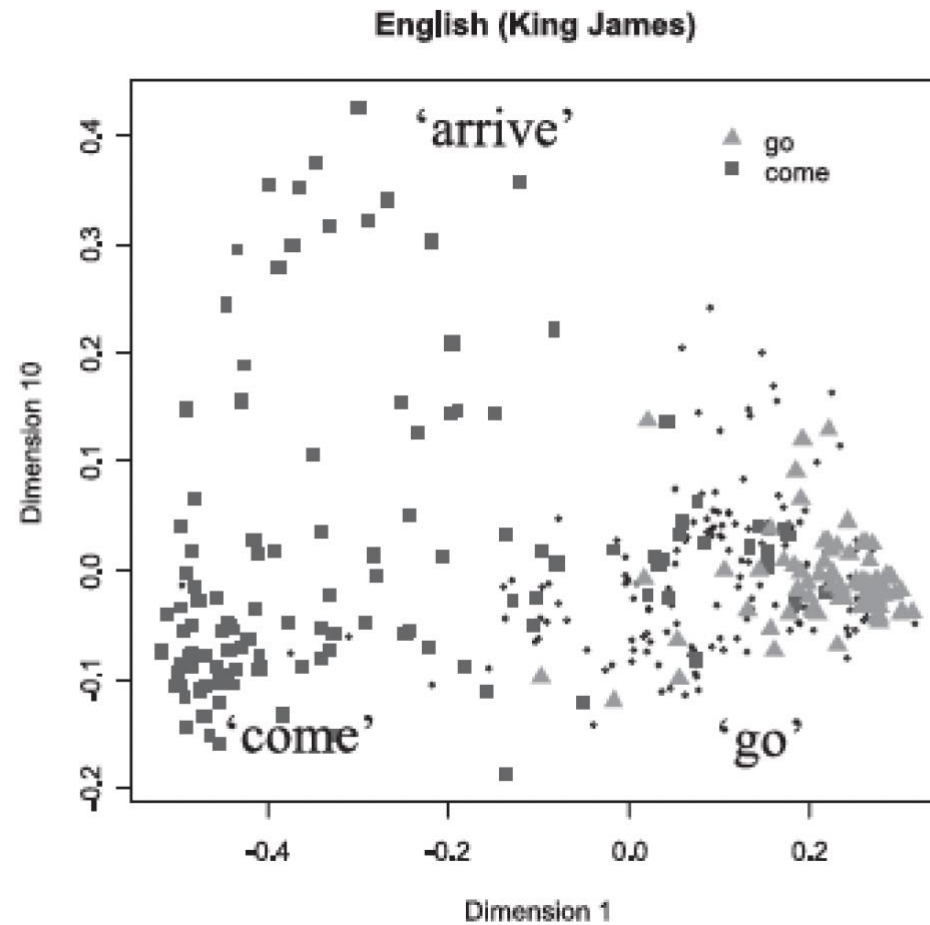
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Proximity maps

- Probabilistic semantic maps are often based on parallel corpora.
- The main unit is a specific context in a parallel corpus (e.g. sentence from the New Testament translated into many languages), not a concept or semantic function.
- The relationships between contexts are represented as distances in MDS, not links.
- The distances between contexts are determined by the similarity between the translations (forms) in the documents.

Wälchli & Cysouw (2012): motion verbs



Steps

1. Collect the data (fictitious example)

	Lang1	Lang2	Lang3	Lang4	Lang5
Sit1	bla	qu	da	nina	haha
Sit2	bla	qu	da	nana	hihi
Sit3	bla	qa	ta	nina	hehe

	Lang1	Lang2	Lang3	Lang4	Lang5
Sit1	bla	qu	da	nina	haha
Sit2	bla	qu	da	nana	hihi
Sit3	bla	qa	ta	nina	hehe

Comparative concepts (cf. Haspelmath 2010)

2. Compute the proportion of dissimilar values between the rows (sutations). They will serve as distances.

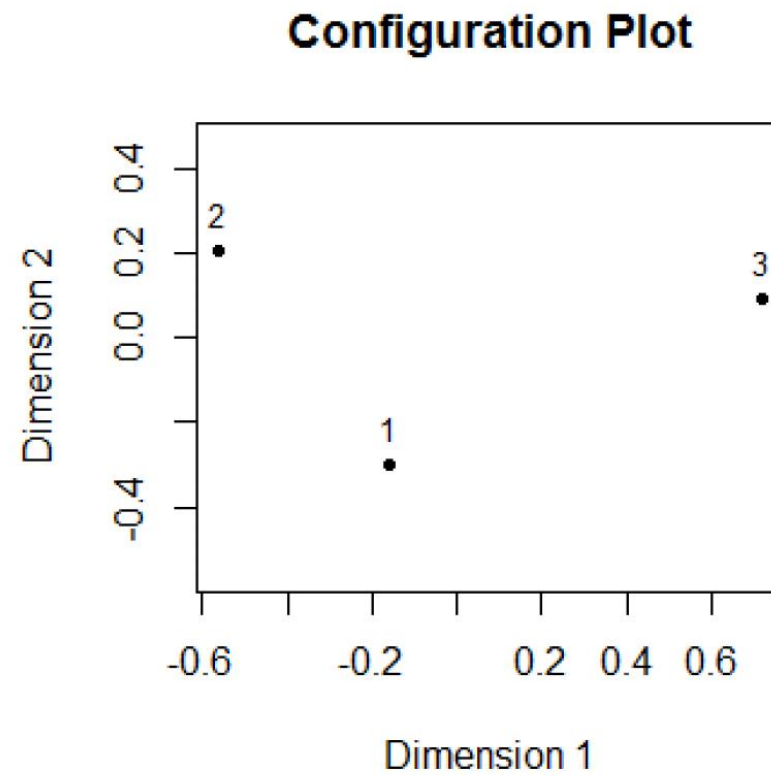
	Lang1	Lang2	Lang3	Lang4	Lang5
Sit1	bla	qu	da	nina	haha
Sit2	bla	qu	da	nana	hihi
Sit3	bla	qa	ta	nina	hehe

$$D(1,2) = 2/5 = 0.4$$

$$D(1,3) = 3/5 = 0.6$$

$$D(2,3) = 4/5 = 0.8$$

3. Perform a MDS analysis.



Intepretation

- The closer two points on the map, the more overlapping constructions they share across the languages.
- Following the isomorphism principle (same function → same form), the corresponding functions/meanings/situations are more semantically similar if more authors of the doculectschose identical constructions to represent these functions/meaning/situations.

Diagnostics

- The less stress (unexplained variance) the better.
- Use a scree plot (stress value) for choosing the number of dimensions.

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ParTy corpus of film subtitles

- Alignment: Jörg Tiedemann's software subalign with manual correction
- Available at <https://github.com/levshina/ParTy-1.0>
- ParTree: Parsed with UD, expanded
Ebert, C., Levshina, N., & Widmer, P. (2023). ParTree - Parallel Treebanks: A multilingual corpus of movie subtitles. (Version 1.0.0) [Data set]. LaRS - Language Repository of Switzerland. <https://doi.org/10.48656/zjzp-gj69>

Languages

- Indo-European
 - Germanic: Dutch, English, German, Norwegian, Swedish
 - Romance: French, Italian, Portuguese, Romanian, Spanish
 - Slavic: Bulgarian, Czech, Polish, Russian, Slovene
- Uralic:
 - Finnic: Estonian, Finnish
 - Ugric: Hungarian

Films for this case study



Examples of analytic causatives

- English: *make*+ Vinf, *let*+ Vinf, *have*+ Vinf, *cause*+ *to*-Vinf
- German: *lassen*+ Vinf
- Dutch: *laten*+ Vinf, *doen*+ Vinf
- Russian: *zastavljat'* “force” + Vinf, *davat'* “give” + Vinf
- French: *faire*+ Vinf, *laisser*+ Vinf
- Romanian: *face*+ *să*+ Vsubj, *lasă*+ *să*+ Vsubj

Examples

- Context A

- ENG: And we make them do it... ...or we kill them. **make**
- ITA: E glielo facciamo fare ... o lo uccidiamo. **fare**
- CZE: Donutíme je to udělat, nebo je zabijeme. **donutit**

- Context B

- ENG: Pick up someone my height and build and make them believe it is me. **make**
- ITA: Individua una della mia corporatura e fa credere loro che sia io. **fare**
- CZE: Vyber někoho, kdo je mi podobn. a přesvědč je, že jsem to já. **NA**